

ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

MAXIMUM LIKELIHOOD ESTIMATION

SANJOY DASGUPTA, PROFESSOR

UC San Diego

COMPUTER SCIENCE & ENGINEERING
HALICIOĞLU DATA SCIENCE INSTITUTE

Maximum Likelihood Estimation

Let $\mathcal{P} = \{P_\theta: \theta \in \Theta\}$ be a class of probability distributions (Gaussians, Poissons, etc).

Maximum likelihood principle: pick the $\theta \in \Theta$ that makes the data maximally likely, that is, maximizes $\Pr(\text{data}|\theta) = P_\theta(\text{data})$.

Three steps:

- 1 Write down an expression for the **likelihood**, $\Pr(\text{data}|\theta)$.
- 2 Maximizing this is the same as maximizing its log, the **log-likelihood**.
- 3 Solve for the maximum-likelihood parameter θ .

Maximum Likelihood Estimation of the Poisson

$\mathcal{P} = \{\text{Poisson}(\lambda): \lambda > 0\}$. We observe $x_1, \dots, x_n$.

- Write down an expression for the **likelihood**, $\Pr(\text{data}|\lambda)$
- Maximizing this is the same as maximizing its log, the log-likelihood.
- Solve for the maximum-likelihood parameter λ .

Maximum Likelihood Estimation of the Normal

You see n data points $x_1, \dots, x_n \in \mathbb{R}$, and want to fit a Gaussian $N(\mu, \sigma^2)$ to them.

- Maximum likelihood: pick μ, σ to maximize

$$\Pr(\text{data}|\mu, \sigma^2) = \prod_{i=1}^n \left(\frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right) \right)$$

- Work with the log, since it makes things easier:

$$\text{LL}(\mu, \sigma^2) = \frac{n}{2} \ln \frac{1}{2\pi\sigma^2} - \sum_{i=1}^n \frac{(x_i - \mu)^2}{2\sigma^2}$$

- Setting the derivatives to zero, we get

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i \quad \sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$$

These are simply the empirical mean and variance.